

# TuniCars+

## Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

20260813\_184324\_c2a886

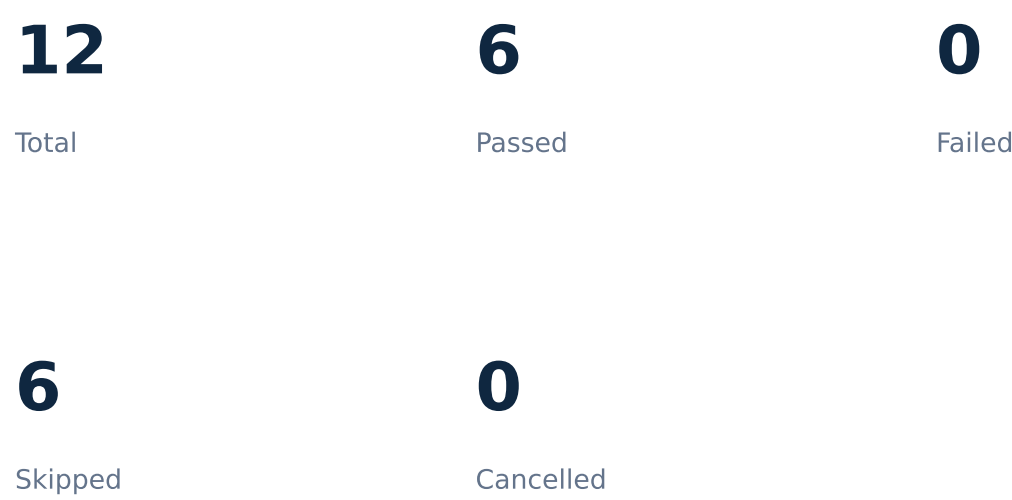
Date

2026-08-13T17:43:24.806625+00:00

# Executive Summary

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Overall measured availability: 71.89%

During this campaign, 6 of 12 scenarios passed, 0 failed and 6 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

# Test Environment

Architecture and technologies

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## Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

# Test Scenarios

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| Scenario          | kubernetes | swarm |
|-------------------|------------|-------|
| container-kill    | 1          | 1     |
| cpu               | 1          | 1     |
| latency           | 1          | 1     |
| memory            | 1          | 1     |
| network-partition | 1          | 1     |
| node-failure      | 1          | 1     |

# Full Results Table

| Scenario          | Platform   | Detection | Recovery | Required | Success      | Avg ms       | P95 ms       | HTTP Fail | Avail. %     | Status  | Skip Reason                                                                                                                                            | Error Message |
|-------------------|------------|-----------|----------|----------|--------------|--------------|--------------|-----------|--------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| container-kill    | swarm      | 1.562     | 10.703   | True     | True         | 7.878        | 16.573       | 10        | 9.09         | PASS    | N/A                                                                                                                                                    | N/A           |
| node-failure      | swarm      | N/A       | N/A      | N/A      | N/A          | Not Measured | Not Measured | N/A       | Not Measured | SKIPPED | Single-node Docker Swarm cannot demonstrate failover routes require authentication, but have no RESILIENCE_API_TOKEN                                   | N/A           |
| cpu               | swarm      | N/A       | N/A      | N/A      | Not Required | 11.098       | 18.459       | N/A       | 100.0        | PASS    | Single-node Docker Swarm cannot demonstrate failover routes require authentication, but have no RESILIENCE_API_TOKEN                                   | N/A           |
| memory            | swarm      | N/A       | N/A      | N/A      | Not Required | 12.7         | 19.322       | N/A       | 100.0        | PASS    | Single-node Docker Swarm cannot demonstrate failover routes require authentication, but have no RESILIENCE_API_TOKEN                                   | N/A           |
| network-partition | swarm      | N/A       | N/A      | N/A      | N/A          | Not Measured | Not Measured | N/A       | Not Measured | SKIPPED | Single-node Docker Swarm cannot demonstrate failover routes require authentication, but have no RESILIENCE_API_TOKEN                                   | N/A           |
| latency           | swarm      | N/A       | N/A      | N/A      | N/A          | Not Measured | Not Measured | N/A       | Not Measured | SKIPPED | Single-node Docker Swarm cannot demonstrate failover routes require authentication, but have no RESILIENCE_API_TOKEN                                   | N/A           |
| container-kill    | kubernetes | 4.734     | 17.469   | True     | True         | 1365.24      | 2045.227     | 7         | 22.22        | PASS    | N/A                                                                                                                                                    | N/A           |
| node-failure      | kubernetes | N/A       | N/A      | N/A      | N/A          | Not Measured | Not Measured | N/A       | Not Measured | SKIPPED | Multi-node kube inter-node CNI/DNS does not provide a safe backend to restore failover routes require authentication, but have no RESILIENCE_API_TOKEN | N/A           |
| cpu               | kubernetes | N/A       | N/A      | N/A      | Not Required | 16.32        | 21.052       | N/A       | 100.0        | PASS    | Multi-node kube inter-node CNI/DNS does not provide a safe backend to restore failover routes require authentication, but have no RESILIENCE_API_TOKEN | N/A           |
| memory            | kubernetes | N/A       | N/A      | N/A      | Not Required | 15.828       | 20.261       | N/A       | 100.0        | PASS    | Multi-node kube inter-node CNI/DNS does not provide a safe backend to restore failover routes require authentication, but have no RESILIENCE_API_TOKEN | N/A           |
| network-partition | kubernetes | N/A       | N/A      | N/A      | N/A          | Not Measured | Not Measured | N/A       | Not Measured | SKIPPED | Multi-node kube inter-node CNI/DNS does not provide a safe backend to restore failover routes require authentication, but have no RESILIENCE_API_TOKEN | N/A           |
| latency           | kubernetes | N/A       | N/A      | N/A      | N/A          | Not Measured | Not Measured | N/A       | Not Measured | SKIPPED | Multi-node kube inter-node CNI/DNS does not provide a safe backend to restore failover routes require authentication, but have no RESILIENCE_API_TOKEN | N/A           |

# swarm — container-kill

State: Pass  
Detection Time (s): 1.562  
Recovery Time (s): 10.703

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HTTP Success: N/A  
HTTP Failures: 10  
Availability (%): 9.09  
CPU (%): N/A  
Memory (MB): N/A

## Recorded interpretation

No error or limitation recorded.

# swarm — node-failure

Status: SKIPPED  
Detection Time (s): N/A

Recovery Time (s): N/A

---

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

## Recorded interpretation

Single-node Docker Swarm cannot demonstrate real node rescheduling.

# swarm — cpu

Status: Pass  
Detection Time (s): N/A

Recovery Time (s): N/A

---

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

## Recorded interpretation

No error or limitation recorded.



# swarm — memory

State: Pass  
Detection Time (s): N/A

Recovery Time (s): N/A

---

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

## Recorded interpretation

No error or limitation recorded.

# swarm — network-partition

Status: SKIPPED

Detection Time (s): N/A

Recovery Time (s): N/A

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HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

## Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE\_API\_TOKEN nor RESILIENCE\_TEST\_IDENTIFIER/RESILIENCE\_TEST\_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

# swarm — latency

State: SKIPPED  
Detection Time (s): N/A

Recovery Time (s): N/A

---

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

## Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE\_API\_TOKEN nor RESILIENCE\_TEST\_IDENTIFIER/RESILIENCE\_TEST\_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

# kubernetes — container-kill

State: **Pass**  
Detection Time (s): 4.734  
Recovery Time (s): 17.469

---

HTTP Success: N/A

HTTP Failures: 7

Availability (%): 22.22

CPU (%): N/A

Memory (MB): N/A

## Recorded interpretation

No error or limitation recorded.

# kubernetes — node-failure

Status: SKIPPED  
Detection Time (s): N/A

Recovery Time (s): N/A

---

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

## Recorded interpretation

Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

# kubernetes — cpu

State: Pass

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

## Recorded interpretation

No error or limitation recorded.

# kubernetes — memory

Status: Pass  
Detection Time (s): N/A  
Recovery Time (s): N/A  
HTTP Success: N/A  
HTTP Failures: N/A  
Availability (%): 100.0  
CPU (%): N/A  
Memory (MB): N/A

## Recorded interpretation

No error or limitation recorded.

# kubernetes — network-partition

Status: SKIPPED  
Detection Time (s): N/A

Recovery Time (s): N/A

---

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

## Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE\_API\_TOKEN nor RESILIENCE\_TEST\_IDENTIFIER/RESILIENCE\_TEST\_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401



# kubernetes — latency

Status: SKIPPED  
Detection Time (s): N/A

Recovery Time (s): N/A

---

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

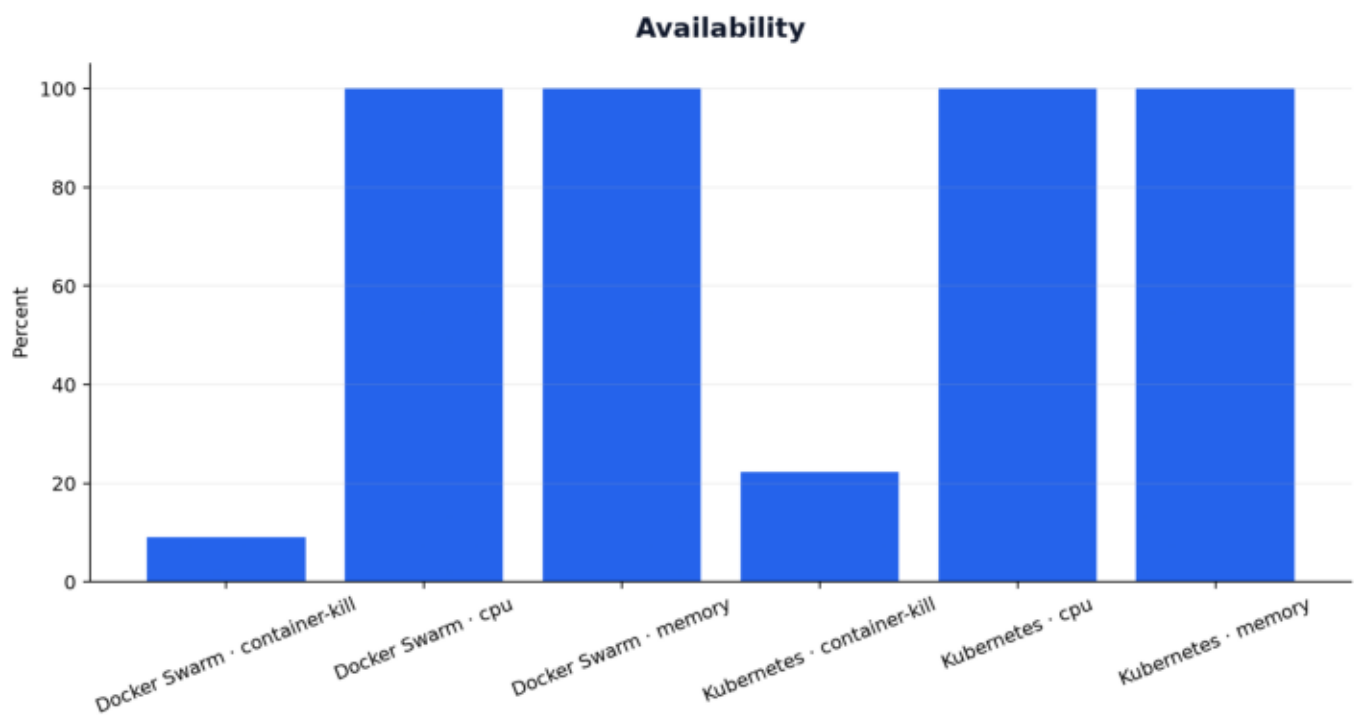
## Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE\_API\_TOKEN nor RESILIENCE\_TEST\_IDENTIFIER/RESILIENCE\_TEST\_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

# Availability

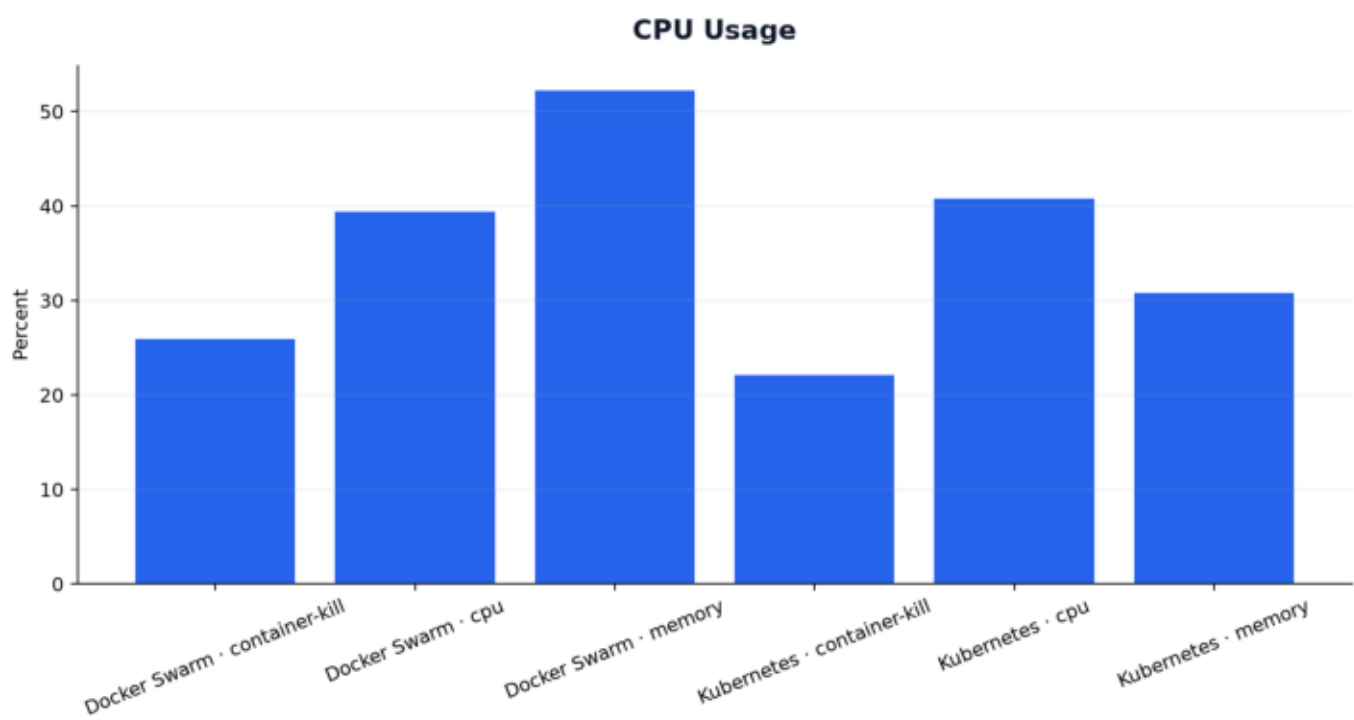
Current-session measured data

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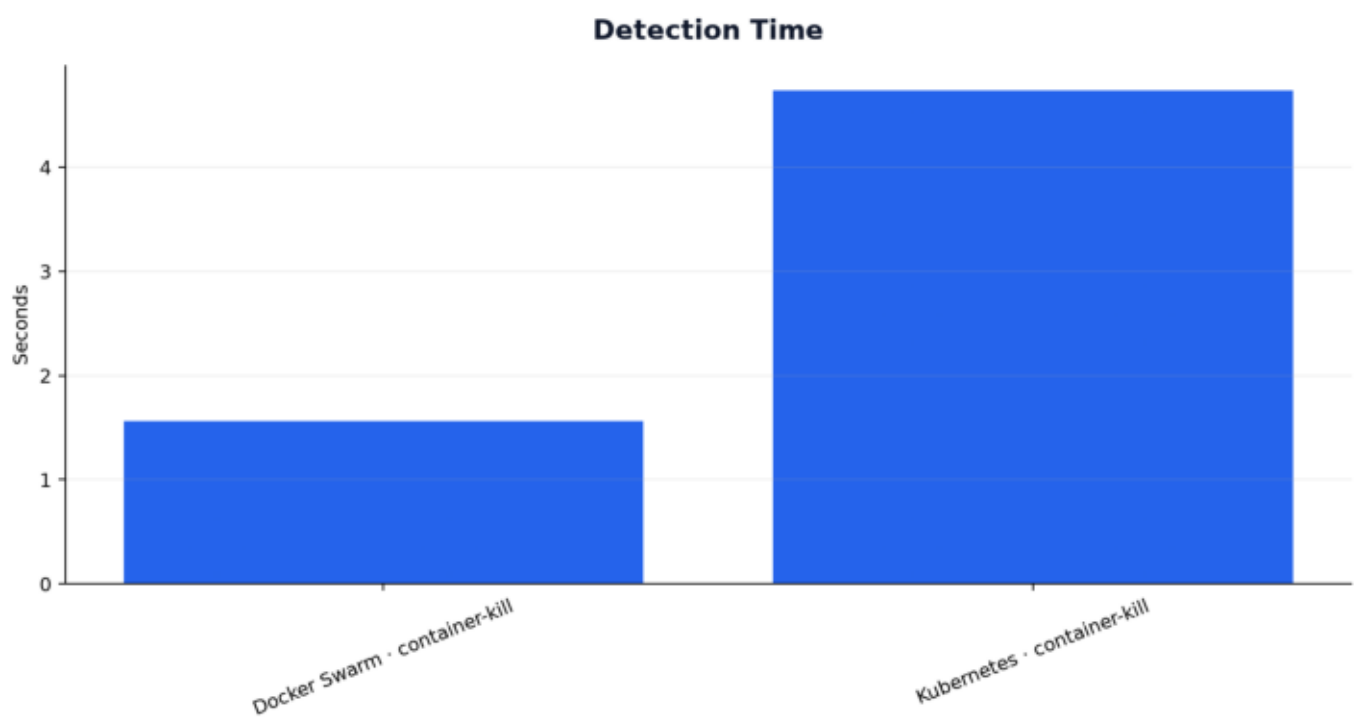
# Cpu Usage

Current-session measured data



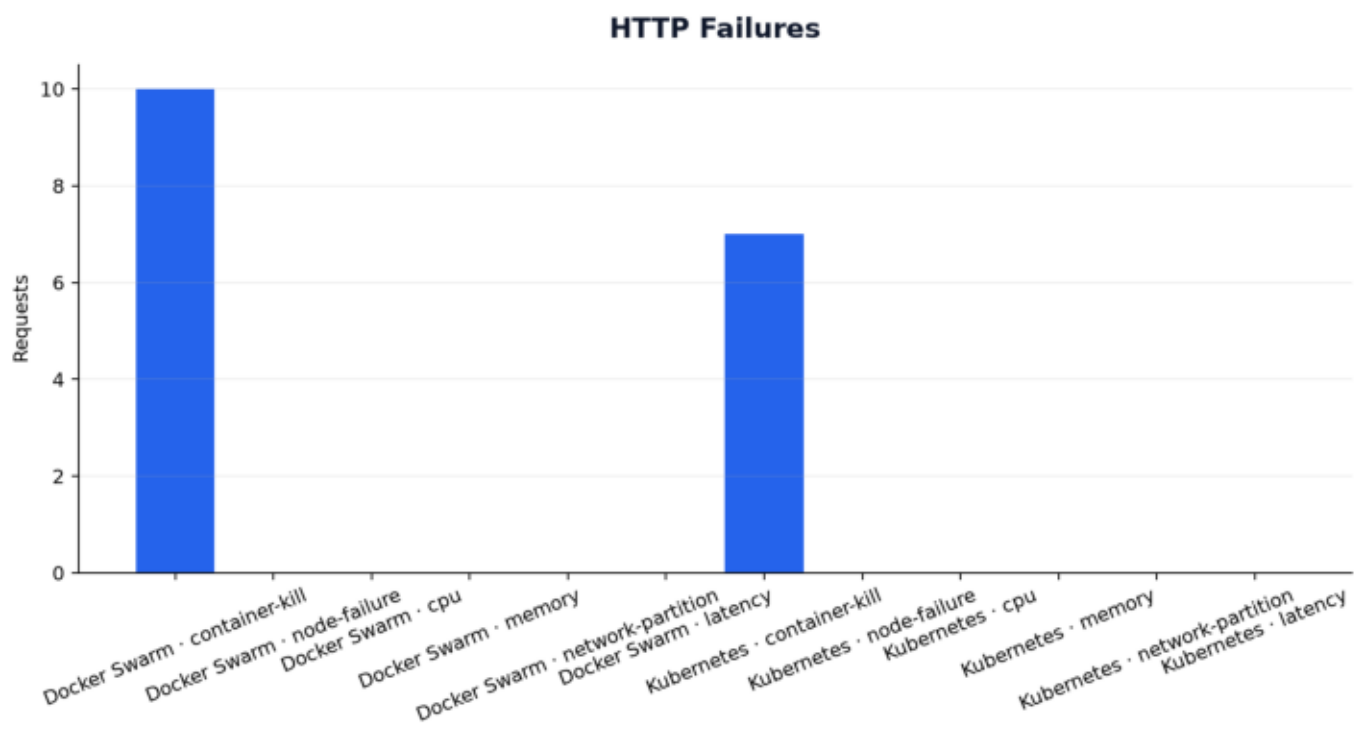
# Detection Time

Current-session measured data



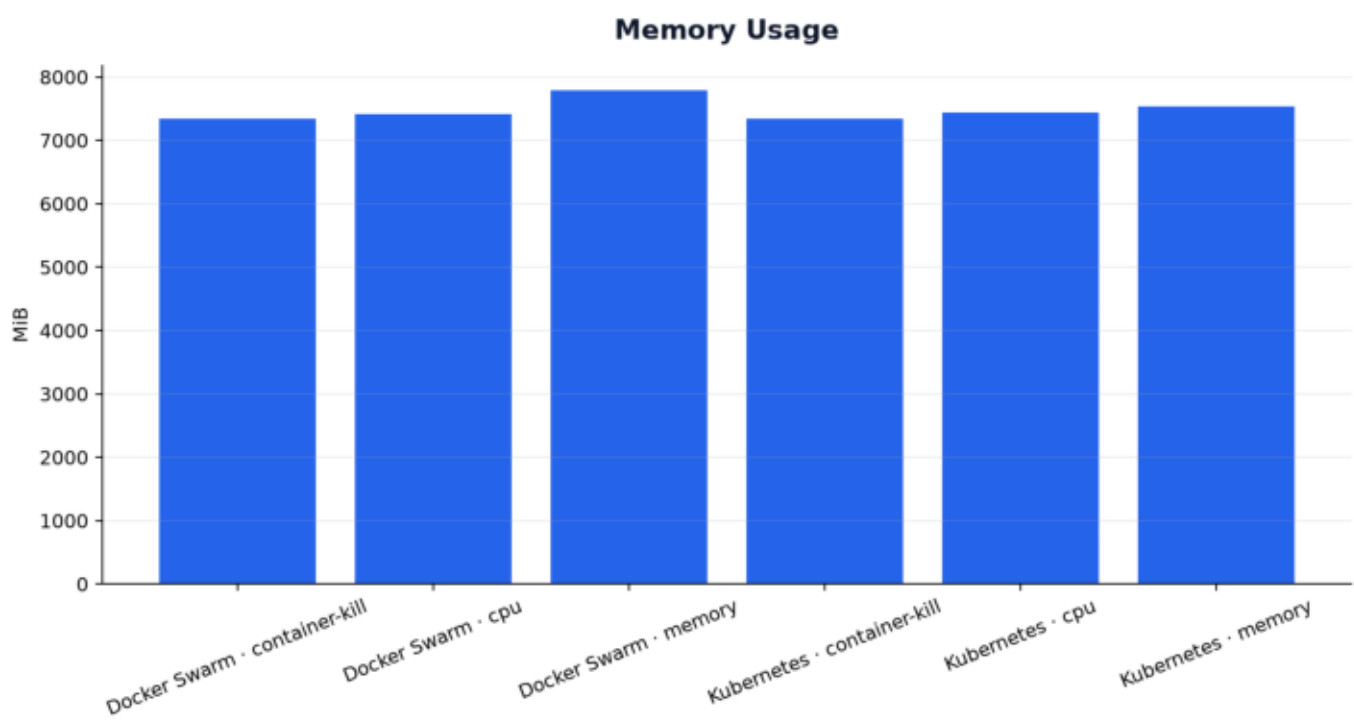
# Http Failures

Current-session measured data



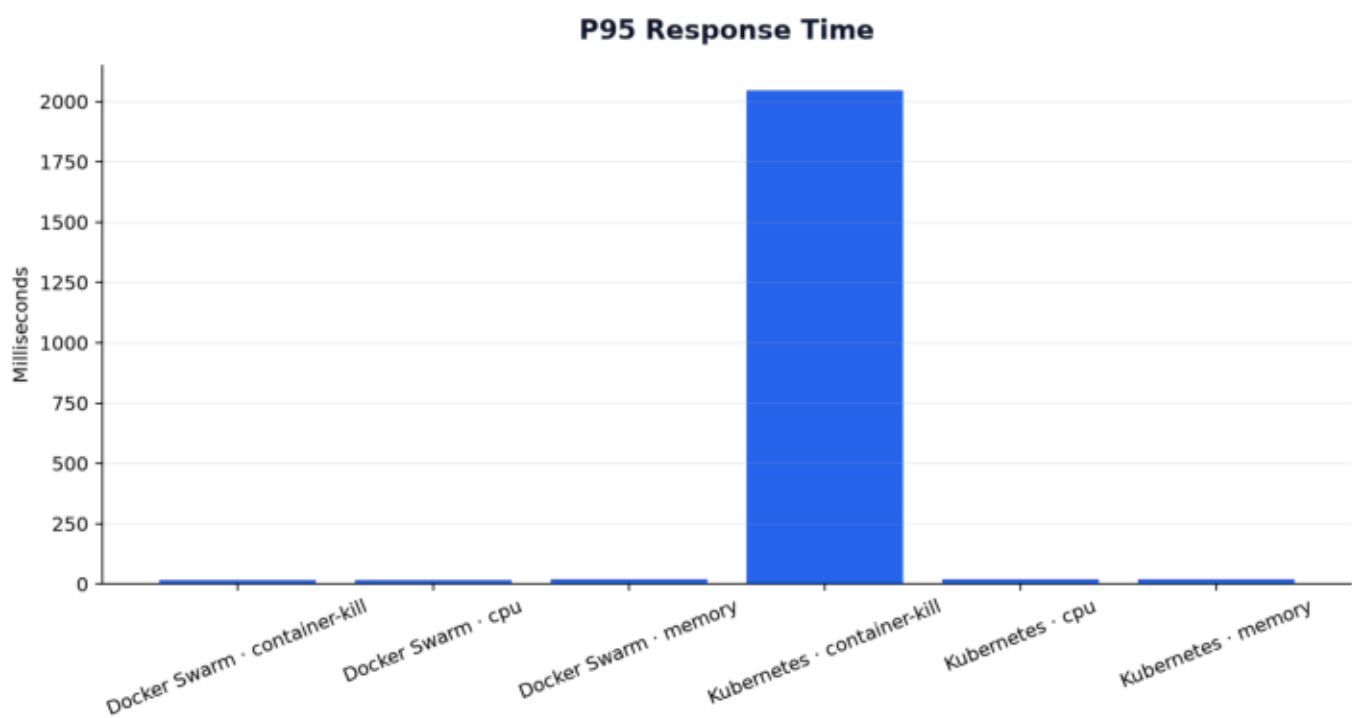
# Memory Usage

Current-session measured data



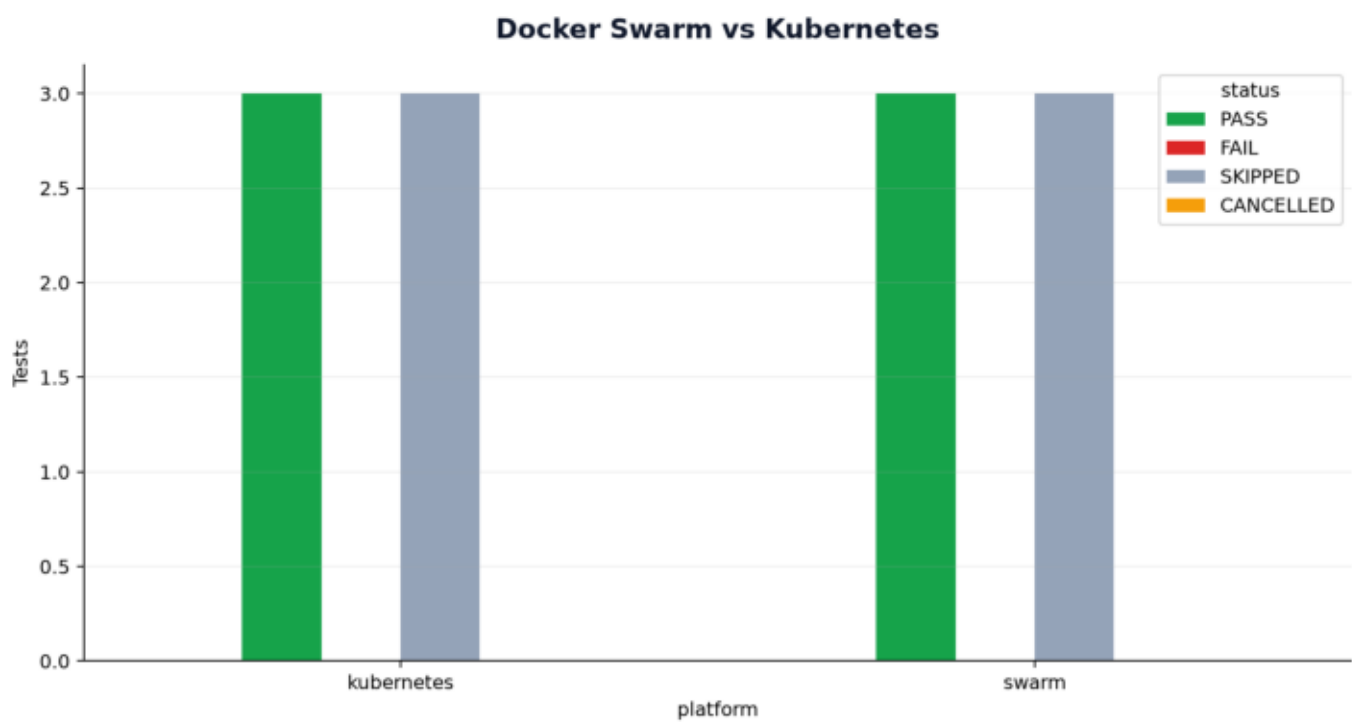
# P95 Response Time

Current-session measured data



# Platform Comparison

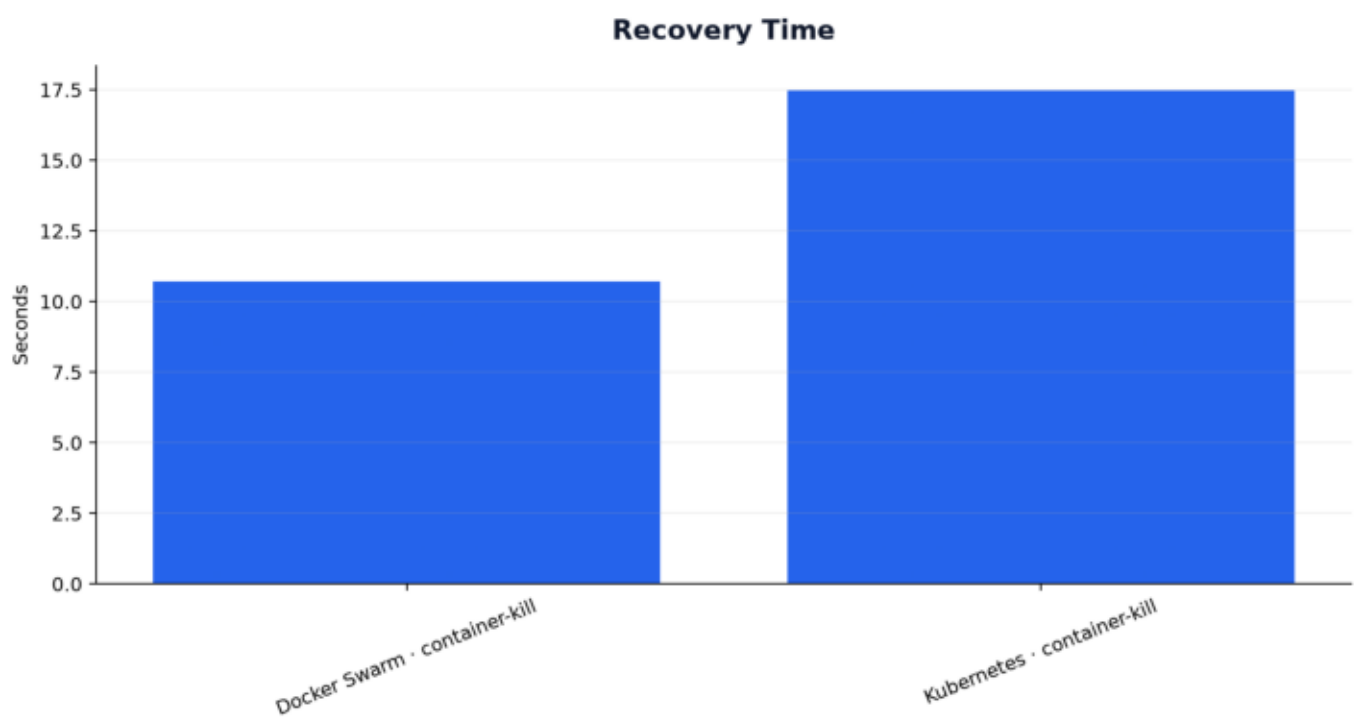
Current-session measured data





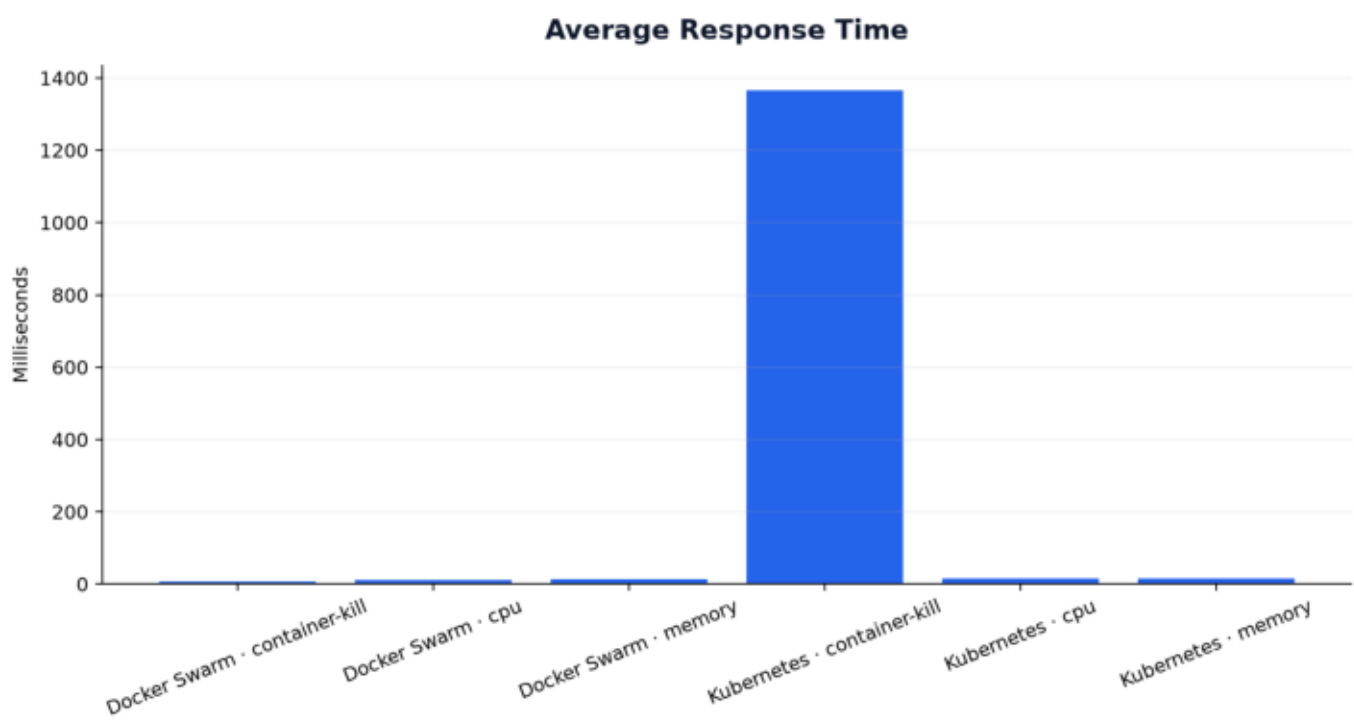
# Recovery Time

Current-session measured data



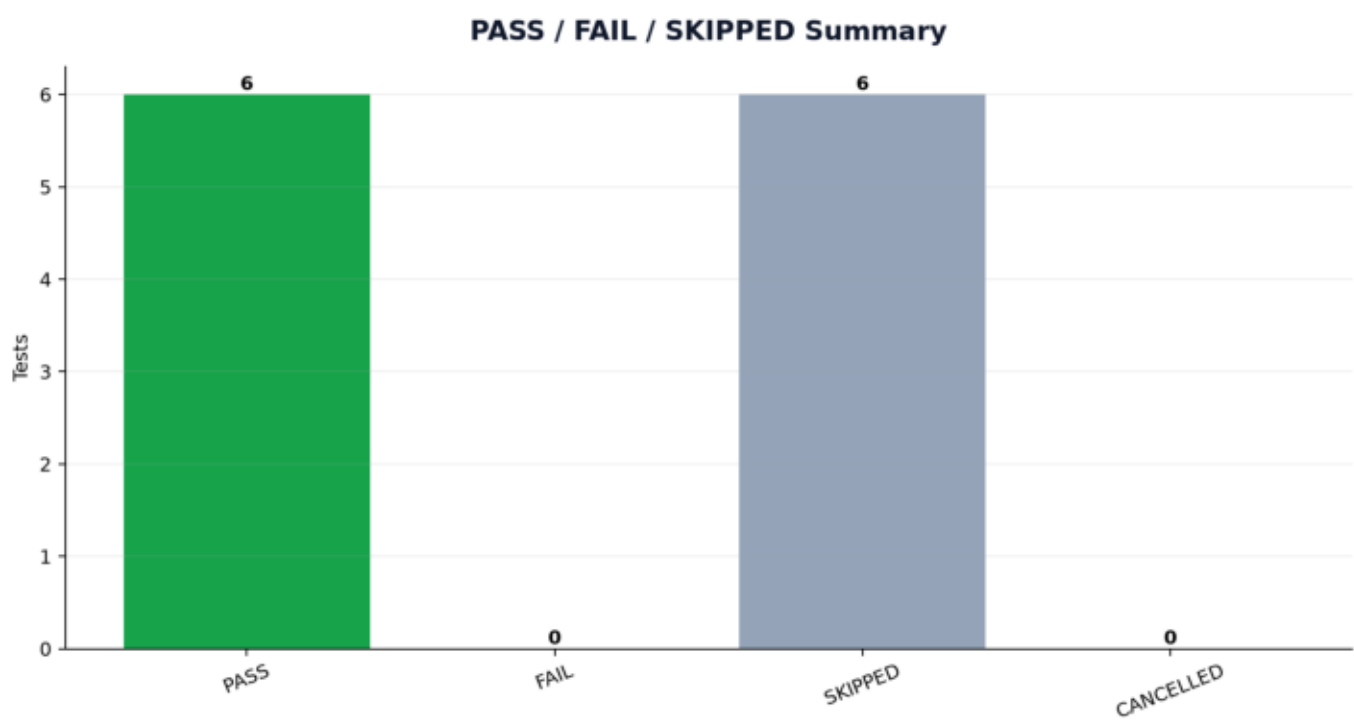
# Response Time

Current-session measured data



# Status Summary

Current-session measured data



# Failures and Errors

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| Platform | Scenario | Status | Error                 |
|----------|----------|--------|-----------------------|
| —        | —        | —      | No failures recorded. |

# Skipped Tests

| Platform   | Scenario          | Exact reason                                                                                                                                                                                                                                                                                                  |
|------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| swarm      | node-failure      | Single-node Docker Swarm cannot demonstrate real node rescheduling. Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401 |
| swarm      | network-partition | CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401                                                                                                                                                                |
| swarm      | latency           | CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401                                                                                                                                                                |
| kubernetes | node-failure      | Minikube inter-node CN/DNS does not provide Database probe to PostgreSQL authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401                          |
| kubernetes | network-partition | CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401                                                                                                                                                                |
| kubernetes | latency           | CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401                                                                                                                                                                |

# Swarm vs Kubernetes Comparison

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| Platform   | PASS | SKIPPED |
|------------|------|---------|
| kubernetes | 3    | 3       |
| swarm      | 3    | 3       |

# Final Conclusion

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During this campaign, 6 of 12 scenarios passed, 0 failed and 6 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.